CST8248 – Emerging Technologies

Term Project

Purpose: The purpose of this project is to capstone learning throughout your experiences in the CST program. You will need to apply skills from each of your courses up to this point.

This project places you in the role of a Managed Service Provider (MSP) working as a team to design a network, manage goals, complete objectives and maintain timelines. You will need to work as a team to achieve these objectives. Communication between team members will be vital for team success. Documentation and organization will also play major roles over the course of this project.

Assignments in this course have been designed to assist you with planning, tracking, and completing objectives, communicating with team members, and reflecting on the product.

Teamwork is essential for producing a quality product.

******Your Lab instructor can take the role of client/employer if you have questions or require clarification about objectives.**

******Your lab professor can offer guidance and suggestions if you require assistance. Consult with your lab instructor early if you require assistance with a scope for value added.**

Background

Sites:

Site 1 will house the MSP Private Equipment. This equipment is located in T114a.

Site 2 will house the MSP Public Cloud Equipment. This equipment is located in T120

Managed Service Provider

You group represents the MSP. You are taksed with providing service to each company based on their individual needs

Company 1

Founded in 2024 and headquartered in **Toronto, Canada**, **Lumora Impact Group** is a startup organization committed to delivering sustainable, measurable impact across health, education, and economic development. Our work focuses on building partnerships, leveraging data, and supporting underserved communities through high-impact, technology-enabled programs.

As a newly launched organization with **no legacy infrastructure**, we have the opportunity to architect a **modern, secure, and compliant IT environment** from the ground up. We are committed to doing this the right way—by partnering with a **Managed Service Provider (MSP)** that can help us design and manage a professional-grade environment that aligns with Canadian data standards and global best practices.

Why We're Seeking a Managed Service Provider (MSP)

Rather than hiring internal IT staff during our early growth phase, we are seeking a long-term MSP partner who can deliver end-to-end support—planning, implementation, maintenance, and strategic scaling.

We prefer to work with **licensed, subscription-based platforms** that offer:

- Predictable, transparent pricing
- Professional support and updates
- Long-term vendor accountability
- Compliance with **Canadian data residency** and privacy regulations.

Our Infrastructure Requirements

We are looking to implement a **hybrid cloud infrastructure** that balances flexibility, scalability, and control. The MSP should help us deploy and support the following components:

🔧 **Core Infrastructure**

- **Public and private cloud integration**
- **Active Directory**, **DNS**, and **DHCP** setup and management
- **Centralized storage**, with potential **SAN (Storage Area Network)** or equivalent solutions
- **Enterprise-grade backup and disaster recovery**, including automated, encrypted offsite/cloud replication

📂 **Collaboration & Productivity**

- **Secure file sharing and collaboration tools**.
- Centralized **identity and access management** with role-based controls

🔐 **Security & Compliance**

- **Endpoint protection**, patch management, and monitoring
- **Firewall, VPN**, and secure remote access configuration

👨💻 **Workstation & User Support**

- **Device provisioning and remote management** for a distributed team
- **User onboarding/offboarding**, and system access

Our Goals

We want to build a technology environment that is:

- **Secure and scalable** from day one
- **Fully supported** without needing internal IT staff
- **Standards-compliant** with Canadian regulations
- Flexible enough to support both remote and physical teams across Canada and beyond

We are seeking a **strategic technology partner**, not just a vendor. Our goal is to establish a long-term relationship with an MSP who understands the unique needs of a mission-driven, high-compliance, growth-stage organization.

Company 2

Organization Backstory: ClearRoots Foundation

Launched in early 2024, the **ClearRoots Foundation** is a startup non-governmental organization dedicated to promoting grassroots environmental education and sustainable farming practices across rural communities in South Asia and East Africa.

Though we are early in our journey, our team is made up of experienced development professionals, educators, and researchers who have worked in international NGOs, rural outreach programs, and policy advocacy. Now, we're building something of our own—lean, impact-focused, and rooted in transparency and collaboration.

As a new organization, we are in the **process of establishing our digital foundation from the ground up**. We have no legacy infrastructure or systems to migrate, which gives us the opportunity to build a **modern IT environment** that reflects our core values: accessibility, sustainability, and open knowledge.

From the start, we've committed to:

- Prioritizing **open-source and license-free software** where possible.
- Minimizing vendor lock-in and maximizing interoperability
- Operating within a **tight budget** while maintaining high standards of security and data integrity

Why We're Seeking a Managed Service Provider

As a lean organization, we do not currently have in-house IT staff. Instead, we are looking to partner with a **Managed Service Provider** who can **help us design, deploy, and maintain** a scalable and cost-efficient IT infrastructure tailored to NGO needs.

What We're Looking For:

We are seeking an MSP that can:

- Recommend and deploy **open-source tools** for file sharing, team communication and collaboration.
- Set up and manage a **hybrid cloud environment**.
- Provide **ongoing system monitoring, patching, backups**, and **security best practices**
- Offer **help desk support**..
- Help us plan for **future scalability**, including onboarding staff, launching regional hubs, and integrating donor reporting tools

We are especially interested in working with partners who understand the unique needs of NGOs—cost constraints, donor reporting requirements, and the importance of open access to information. If your team is passionate about making technology accessible and ethical, we're ready to build with you.

Objectives:

The items below represent your client's/employer's must haves

MSP Requirements

- Connect cables correctly and manage cabling neatly.
- Configure IPMI on each Server.
- Configure a Type 1 hypervisor of your choice in Site 1.
- Configure a Storage Server and provide storage to your Hypervisor.
- Your hypervisor should have 4 disks 2 for the hypervisor OS and 2 for local VMs
- Your storage server should have 10 disks and should connectoto your hypervisor via SMB <u>or</u> NFS <u>and</u> iSCSI
- Confgure Storage for your server (OS, VMStorage).
- Configure off site storage for backups.
- Create a separate isolated network for Company 1 and Company 2.
- Create a management network Separate from any Company Networking.
- Connect each company's private cloud to its public cloud securely.
- Backup a VM in the Private Hypervisor using Veeam Community Edition.
- Configure one or more Jump Server(s) and ensure that it is loaded with appropriate management tools.
- Backup each company's file shares using Veeam.
- Backup company Client Operating Systems using Veeam Agent Backup.
- Create at least 1 Administrator account <u>per team member.</u> Each Team member should log in with their own account. Accounts should not be shared.
- Create user accounts for each company and ensure proper access based on user profiles.
- Provision for live and cold VM Migratiion for High Availability
- Prepare for automated VM deployments in Proxmox.
- Ensure that users cannot access admin/management tools.
- Firewalls should be enabled on all devices.

- Ensure network connectivity to all devices.
- Ensure that devices have service connectivity.

Company 1 Requirements

- Deploy the following services on the platform(s) of your choice following **best practices** and meet the particular company requests.
    - Web Server in a DMZ
    - DNS Services for local name resolution
    - Separate DNS Services for recursive name resolution
    - Fault Tolerant/Replicated File Server with appropriate access control for company users.
    - Active Directoy or LDAP
    - 2 Client Machines. 1 Linux and 1 Windows
    - Fault Tolerant DHCP
    - iSCSI Target
    - iSCSi initiator
- Client Machines should have access to the file shares
- User accounts should have file share mapped/mounted
- iSCSI traffic should be isolated from user generated traffic.
- The client reserves the right to scale out client machines as they will be onboarding new employees.
- The Windows Client should be accessible via RDP from the college network

Company 2 Requirements

- Deploy the following services on the platform(s) of your choice following **best practices** and meet the particular company requests.
    - Web Server in a DMZ
    - DNS services for local and recursive name resolution
    - Fault Tolerant/Replicated File Server with appropriate access control for company users.
    - Active Directory or LDAP
    - 1 Client Desktop Operating System
    - Fault Tolerant DHCP
    - iSCSI Target
    - iSCSi initiator
- Client Machines should have access to the file shares
- User accounts should have file share mapped/mounted
- iSCSI traffic should be isolated from user generated traffic.
- The client reserves the right to scale out client machines as they will be onboarding new employees.
- The Linux Client should be accessible via SSH from the college network.

Remote Access must be enabled for all machines (RDP, SSH, VNC, etc.) the use of Virtual console is not acceptable**

Value Added:

This section represents your team's ability to exceed your client/employer expectations
(What makes your team stand out above the competition)

For full marks, each team member must deploy at least one value added service

- Deploy platforms and services that will assist in managing, securing, monitoring and/or administrating the network.
- Tech you always wanted to try, and think might be useful to your client
- Emerging Technologies you are interested in experimenting with.

Value-Added Services can be related/integrated with the whole team or individual efforts. This aspect of the project is intended to be a research project. Your Value-Added element cannot be any technology that has been taught previously in the CST Program. Services previously studied in CST may be used as component parts of Value Added. Consult with your lab professor to confirm the scope of your value added.

Equipment:

- Access to a hypervisor
  - CIA Cluster
  - 2x DL380 Servers
  - Student Laptop
  - Lab Workstation
- ISO's of your choice of Linux or Windows platforms
  - ISO's of windows can be downloaded through the Azure portal -> Education -> Software
- Veeam Community Edition can be downloaded from Veeam
  - https://www.veeam.com/virtual-machine-backup-solution-free.html

Rubric:

70% of clients/employer's must haves

30% Value Added